Part of the UPPP Series

April 2025

UTAH PRISON PROJECTIONS PROJECT: PROCESS & METHODOLOGY

Issue: This document outlines the process and statistical approach used to forecast Utah's prison population. Our framework is designed to be dynamic and adaptable, allowing for the integration of new data and the accommodation of shifting trends.

Overview

The Utah Prison Projection Project (UPPP) is a collaborative effort between the Utah Department of Corrections (UDC) and the Utah Commission on Criminal and Juvenile Justice (CCJJ). This document provides an overview of the process and statistical approach employed to forecast Utah's prison population. Future projection efforts will be designed to adapt to changes in policies and legislation, and to allow for deeper analysis of key sub-populations. The full UPPP project plan is available here.

Process & Data

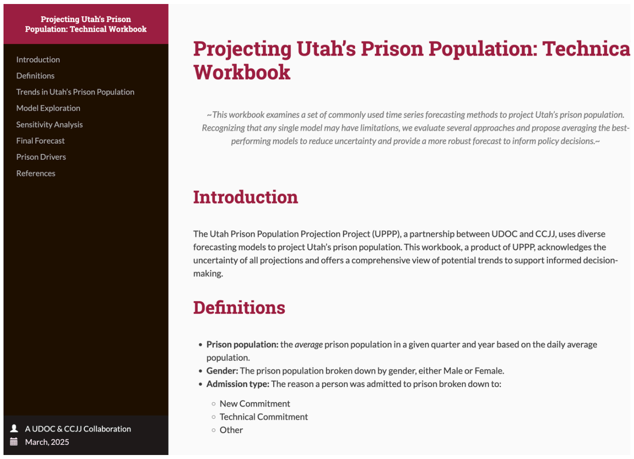
Utah developed its prison population forecasting framework through a comprehensive review of the national literature and consultations with local expertise and selected states to understand their successes and challenges in prison projection work. This review guided the exploration of commonly used forecasting methods to be analyzed on Utah data.

The analytical process began with a descriptive analysis to identify trends, seasonal variations, and irregular patterns using historical quarterly prison population data between 2010 and 2024.

The forecasting model presented in the **UPPP Dashboard** follows a two-step, hybrid approach: it combines a linear model with structural controls and a residual adjustment using exponential smoothing to account for unexplained variation. These structural controls were incorporated to address the sharp shifts in Utah's prison population following the Justice Reinvestment Initiative (JRI) in 2016 and the COVID-19 pandemic in 2020. Additionally, general population trends and violent crime rates were analyzed to assess their influence on future growth. The projection work is formalized in the *UPPP Technical Workbook* as visualized in Figure 1.

The UPPP methodology was selected through a review of the literature, local expertise and learning from other states.

Figure 1: *The UPPP Technical Workboook*


Projecting Utah's Prison Population: Technical Workbook

Introduction
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A UDOC & CCJJ Collaboration
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Projecting Utah's Prison Population: Technica Workbook

~This workbook examines a set of commonly used time series forecasting methods to project Utah's prison population. Recognizing that any single model may have limitations, we evaluate several approaches and propose averaging the best-performing models to reduce uncertainty and provide a more robust forecast to inform policy decisions.~

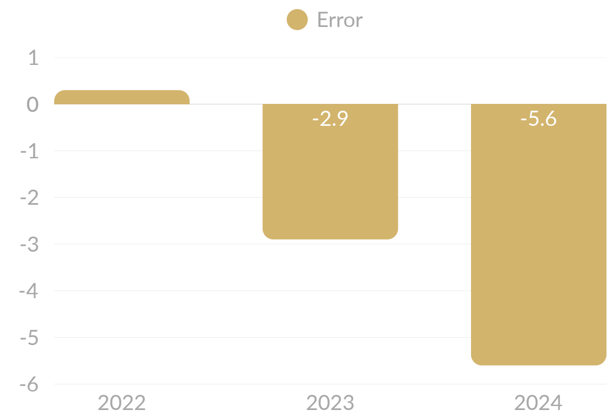
Introduction

The Utah Prison Population Projection Project (UPPP), a partnership between UDOC and CCJJ, uses diverse forecasting models to project Utah's prison population. This workbook, a product of UPPP, acknowledges the uncertainty of all projections and offers a comprehensive view of potential trends to support informed decision-making.

Definitions

- **Prison population:** the *average* prison population in a given quarter and year based on the daily average population.
- **Gender:** The prison population broken down by gender, either Male or Female.
- **Admission type:** The reason a person was admitted to prison broken down to:
  - New Commitment
  - Technical Commitment
  - Other

Figure 2: *Annual average error rate (predicted vs actual as a % of all inmates)*



Training & Validation

Statistical models were trained on historical prison data to evaluate predictive performance using a 80/20 split training and testing data set. Many models demonstrated a tendency to underestimate two-to-three year growth post COVID-19. Uncertainty remains around lasting Covid-19 impacts.

Figure 2 presents the average annual error rate, calculated as the difference between predicted and actual values, divided by the total inmate population. As seen and is commonly observed, the error rate was lowest in short-term forecasts (within 1-year). Model selection prioritized minimizing prediction errors while maintaining simplicity and future applicability. Notably, the 80% and 90% confidence intervals highlight substantial uncertainty around the point estimates.

Ongoing validation improves accuracy for effective policy decisions

Limitations & Updates

Current projections do not account for recent or emerging legislative and policy changes, which will be incorporated in future work. The Technical Workbook provides a framework for updating projections as new data emerges. Ongoing validation and refinement are crucial for projection accuracy and informed policy decisions. Future iterations may enhance forecasting precision with a simulation-based approach.

Acknowledgements

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